

Sample End Semester Question Paper - 2025

Subject: Digital Logic Design | Code: CS203 | Semester: 2

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. State the difference between a latch and a flip-flop.
2. What is the race-around condition?
3. Name two types of sequential circuits.
4. What is a shift register?
5. Define memory word size.
6. Convert $(25)_{10}$ to binary.
7. State one use of 2's complement.
8. What is a K-map?
9. Name two universal gates.
10. How many inputs does an 8:1 multiplexer have?

PART B - Long questions (10 x 8 = 80 Marks)

1. Explain memory units: SRAM, DRAM, ROM and EPROM.
2. Explain the 8085 microprocessor architecture with a block diagram.
3. Explain number system conversions between binary, octal, decimal and hexadecimal with examples.
4. Explain 1's and 2's complement arithmetic with examples.
5. Simplify Boolean expressions using 4-variable and 5-variable K-maps.
6. Explain combinational circuits: adder, subtractor and comparator with logic diagrams.
7. Design an 8:1 multiplexer and a 4:1 demultiplexer with explanations.
8. Explain SR, D, T and JK flip-flops: truth tables and excitation tables.
9. Explain synchronous counters. Design a mod-8 counter.
10. Explain shift registers and their applications.